

Experimental Results

SIFT1M hybrid SQL-vector workload; $k = 10$

Evaluation protocol

- 32 generated hybrid predicates, each paired with a SIFT query vector.
- PRE, POST, GRAPH, and ISECT were evaluated on every query (128 executions).
- Recall@10 is measured against exact top-k results within the predicate-qualified set.
- Latency is end-to-end wall-clock time; vector scans are the HNSW probe budget or exact candidates.

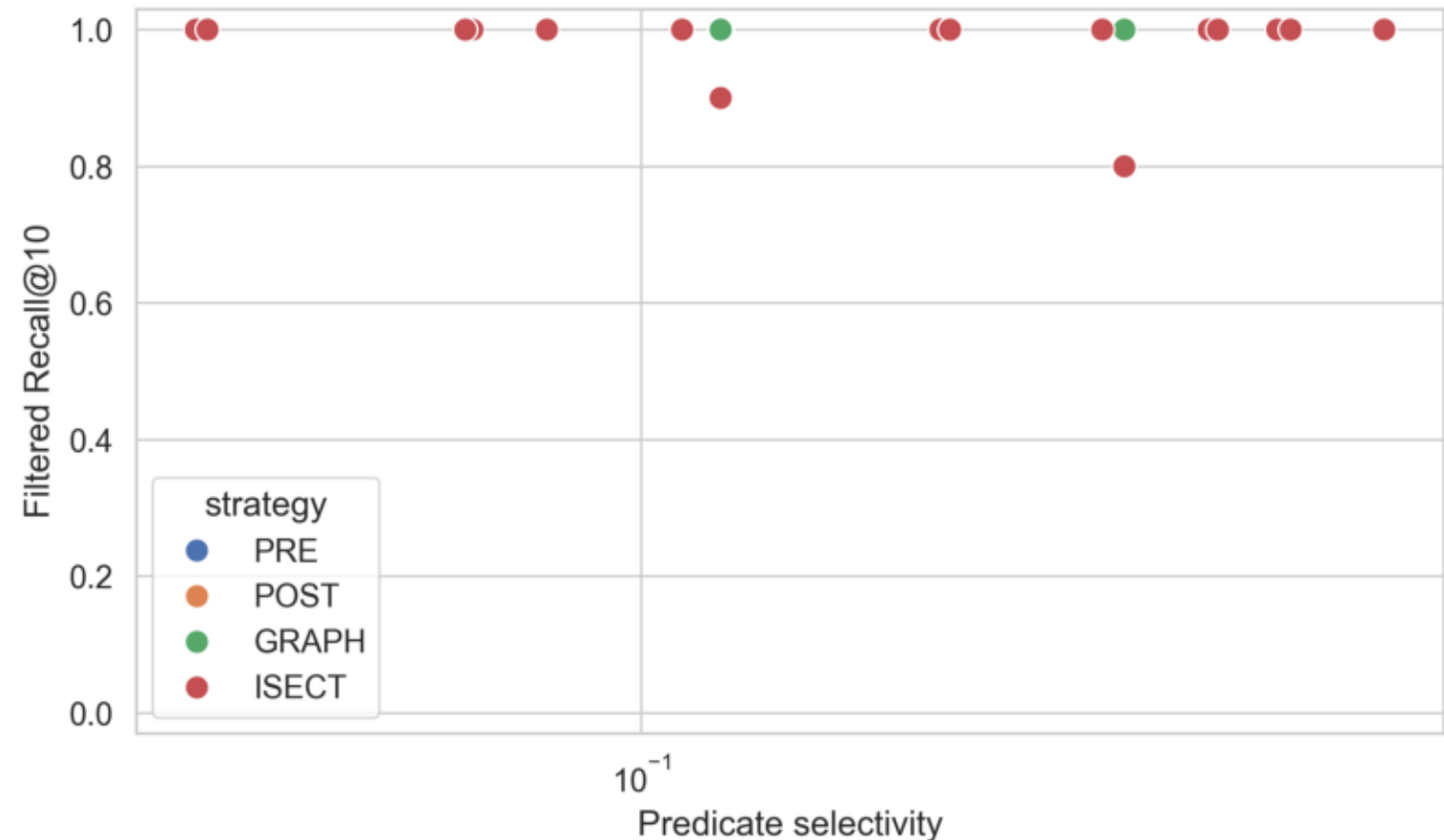
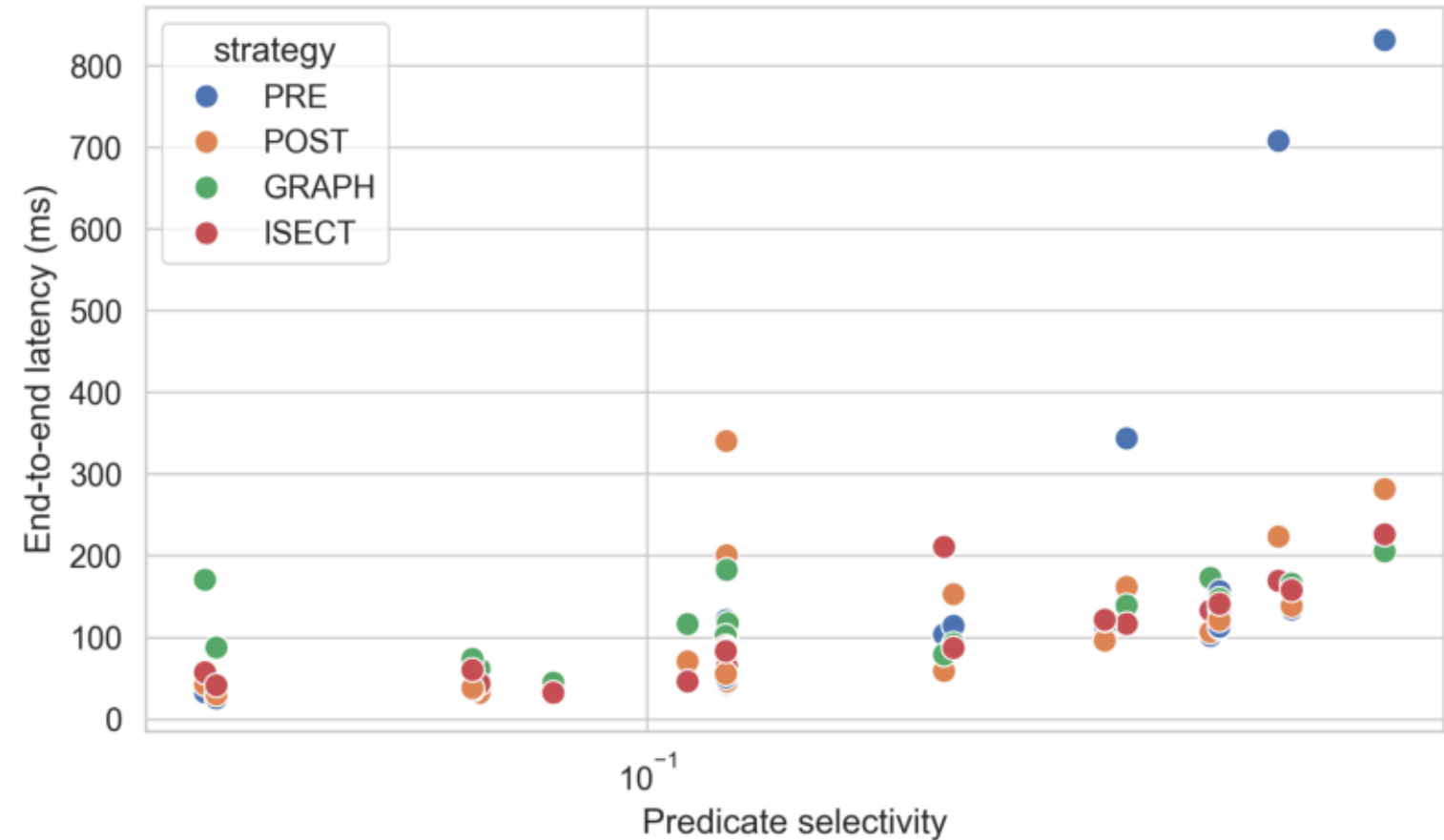
Forced-strategy comparison

Strategy	Mean latency (ms)	Mean Recall@10	Mean vectors scanned
GRAPH	108.56	1.000	233
ISECT	91.77	0.981	233
POST	93.00	0.981	233
PRE	118.79	1.000	220946

Adaptive planner outcome

The calibrated planner selected GRAPH for all 32 workload queries. It achieved mean filtered Recall@10 = 1.000, mean latency = 274.11 ms, and mean probes = 233.

Latency and Recall versus Predicate Selectivity



Planner Decisions and Execution Cost

